

ACE, control and the cost of a proof

Omphalos / Delphyne / miniF2F in Rocq

—55.5%

spending

Accepted bounded-control trade-off;
28 → 25 validation solves.

+3 / 40

Luna ACE increment

Matched development comparison;
encouraging, $p = 0.25$.

38 / 40

Terra during learning

Training generation; later frozen
evaluation uses a different controller.

- **Thesis contribution:** economical, verifiable search through Delphyne.
- **Main question:** which improvements survive the entire path to a checked proof?

Strong control result; modest ACE signal; concrete explanations for lost opportunity.

The questions this update answers

Is ACE implemented?

- **Yes:** Generator, Reflector, Curator and persistent playbook.
- **Local additions:** batched Reducer, checked evidence and budget controls.

Did it improve coverage?

- **Luna:** 24 → 27/40 in the matched validation comparison.
- **Uncertain:** three gains; too little evidence for a supported general gain.

Does a stronger model unlock ACE?

- **Terra solves more** without a book.
- **ACE increment stays small:** +0 with Luna's book, +1 with its own.

What is worth retaining?

- **Budget trade-offs and reliable evidence.**
- **Failed candidates stay optional;** common controller limitations remain.

Separate implementation fidelity, local correctness, coverage and economic value.

Our ACE loop, in one slide

01 Generator

Propose a proof
using the current book



02 Rocq feedback

Execute and check
what actually happened



03 Reflector

Diagnose the attempt;
propose useful lessons

04 Curator

Propose structured
playbook changes



05 Reducer

Select / combine
batched proposals



06 Merge + store

Deterministic update;
hash the next book

- **Training:** the book evolves between batches; no model weights are updated.
- **Evaluation:** the selected book is frozen; each proof has its own resource allowance.
- **Delphyne:** strategies describe choices; policies choose models, search and spending.

The Reducer is a local batching component; the deterministic merge owns the stored state.

Where each ACE element lives in the code

ACE element	Delphyne strategy / query	File to open · then search symbol
Generator	prove_theorem_ace / ProposeProofScriptACE	prove_ace.py
Reflector	reflect_on_trajectory / ReflectOnTrajectory	prove_ace.py
Curator	curate_playbook / CuratePlaybook	prove_ace.py
Reducer	aggregate_curation_deltas / AggregateCurationDeltas	prove_ace.py
Book + merge	Playbook / merge / refine; PlaybookStore.put	ace/ace_playbook.py; ace/ace_store.py
Driver	execute / ACEAdaptStepConfig	experiments/ace/ace_adaptation.py

All paths are relative to examples/omphalos; search by symbol because active code can move lines.

What counts as a solve—and what is capped?

Setting	Dollar allowance / problem	Requests / verifier	Interpretation
Historical X / Luna	\$0.10	32 requests	Older post-crossing stopping
Bounded / Luna	\$0.10	64 / 300 seconds	Conservative pre-request admission
Bounded / Terra	\$1.00	64 / 300 seconds	Same controller; recorded tariffs ×10
Terra training Generator	\$1.00 nominal	32 requests	Evolving book; older stopping rule

- **Qualified solve:** verifier success within the stated cost rule.
- **Bill includes failures;** book preparation is separate.
- **40 problems per X partition;** validation is reused development data.

A nominal allowance, a request reservation and the eventual bill are different quantities.

Development cycles 1/4 · replication and X3

Cycle / change	Observed development result	Conclusion / disposition
Initial replication	ACE = baseline: 32/40 cells ; \$0.5058 vs \$0.4446	No coverage gain; 20 original validation problems × 2 seeds.
Initial ablations / definitions	No-Reflector changes book composition; definitions fix 1/8 cells	No context-collapse replication at this scale; preserve the input fix.
X v2: offline / online	Offline 54/80 = baseline; cold online 56/80 ; warm 50/80	Warm start does not reliably improve results.
X3: batched evidence	X3 57/80 vs baseline 54/80	Modest gain; one-epoch book becomes the reference.
Reflection / rewriting / epochs	No-Reflector 57/80 ; monolithic 55/80 ; X3 3 epochs 53/80	More reflection / training is not reliably better.

For X rows, 80 cells = the same 40 validationX problems at two replicate identifiers.

Development cycles 2/4 · better advice and delivery

Cycle / change	Observed development result	Conclusion / disposition
X4: cited-only credit	55/80 vs X3 57/80	Better bookkeeping; citations do not establish causal usefulness.
Stronger writers / v5	Terra writers 55/80 ; v5 53/80	More capable writing and richer failure evidence do not beat X3.
Rendering v2 vs v3	Cost effect changes sign across v5 and X3 books	Rendering effect depends on the book.
Hints on error / x6 repairs	Triggered 53/80 ; repairs 56 raw / 55 qualified	Some repairs help locally; no reliable large coverage gain.
Stall stopping: k = 4	26/40 , \$0.7943 vs 27/40 , \$0.8870	Spend less with one fewer solve; live and offline estimates differ.
Deterministic digest table	51/80 vs X3 rendering 2 56/80	Simple summaries lose coverage; effect uncertain.

Advice can be correct, cheaper to send, or more traceable without completing more theorems.

Development cycles 3/4 · control and local checks

Cycle / change	Observed development result	Conclusion / disposition
Review development stage	X3 57/80 ; repair books 55 / 54 raw ; baseline 52/80	Repair books do not beat X3; protected stages excluded.
Bounded money + focused	25/40 vs 28/40 ; spending −55.5%	Accepted trade-off ; original no-loss gate failed.
Control-cycle follow-up	28/40 , \$1.0253 vs 25/40 , \$0.6780	Coverage bought back at higher cost; Pareto gate failed.
Polish / downshift / recovery	53/80 each; candidate bill +5.55%	No benchmark win; keep optional.
Applicability / checked change	Adaptive 2/12 useful vs fixed 3/12 ; checked change 1→2/6	Local correctness helps; full-theorem gains unestablished.
Local reasoning-effort probes	Terra low/medium/high 4/6 each; Luna high/xhigh 5/6	More reasoning is not monotonic; check the right local operation.

Local episodes, theorem cells and campaign bills have different denominators.

Development cycles 4/4 · coverage and attribution

Cycle / change	Observed development result	Conclusion / disposition
Model / writer-effort suite	Pilot 18/24 vs 17/24 ; full contenders 19 / 24 vs 27/40	Promising pilot did not become a coverage upgrade.
Structured mechanisms	Effort arm 43/80 vs 53/80 ; S/C/R have 51 parse failures	No promotion; parser confounds mechanism interpretation.
Fixed-book demos / exploration	D/E validation 26 / 25 vs 27/40	No coverage gain; unchanged book.
Four isolated coverage arms	D2/E2/S/H 25 / 24 / 25 / 25 vs 27/40	All cost more on validation; retain the reference.
Attribution / capacity	Own-book increment: Luna +3 , Terra +1 ; training Terra 38/40	Solver strength exceeds demonstrated ACE increment.
13 Sep: resource completion	Small cap 26/40 ; continuation 23/40 vs 27/40	Both cost more; no follow-up selected, no promotion.

Completed cycles are preserved as evidence; the latest losing variants are not the new default.

The strongest result is a budget trade-off

QUALIFIED SOLVES / 40

Older X3  **28/40**

Bounded  **25/40**

WHOLE-PANEL DOLLARS

Older X3  **\$1.5238**

Bounded  **\$0.6780**

- **Coverage:** 28 → 25 / 40.
- **Bill:** \$1.5238 → \$0.6780;
55.5% less.
- **Cost per solve:** \$0.0544 → \$0.0271.
- **Accepted:** three fewer solves for lower cost; no-loss gate failed.

Both use a nominal \$0.10/problem allowance; the controller changes what gets admitted.

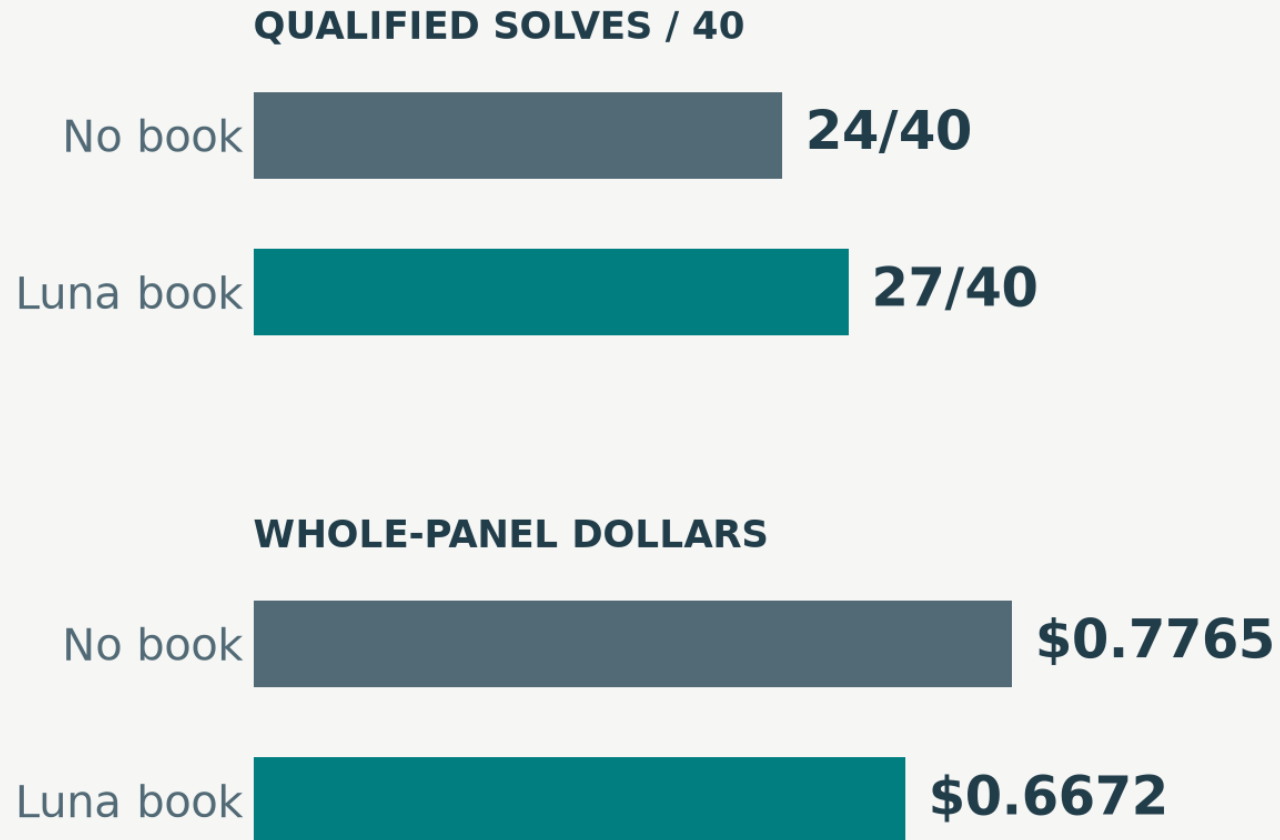
A changed baseline explains part of the confusion

Program	trainX	validationX	Validation bill
Earlier / no book	33/40	27/40	\$0.8870
Bounded / no book	28/40	24/40	\$0.7765
Bounded / Luna book	28/40	27/40	\$0.6672

- **Earlier → current no book:** five training and three validation solves lost.
- **Current no book → ACE:** three validation solves recovered.
- **Current ACE vs earlier program:** same validation count; bill **24.8% lower**.

All use Luna and \$0.10/problem; controller, feedback and runtime lineage still differ.

Luna's book helps here—but the signal is small



- **Question:** does this book help the bounded solver?
- **Observed:** 24 → 27/40; three gains, no losses.
- **Economics:** \$0.7765 → \$0.6672; bill **−14.1%**.
- **Uncertain:** paired $p = 0.25$.

Luna / medium / \$0.10 per problem / 64 requests / 300 verifier seconds.

The full crossover separates solver from book

	No book	Luna book	Terra book
Luna solver \$0.10 cap	24 / 40 \$0.7765	27 / 40 \$0.6672	24 / 40 \$0.7342
Terra solver \$1.00 cap	29 / 40 \$6.6106	29 / 40 \$6.8523	30 / 40 \$6.4513

- **Same Luna book:** +3 solves for Luna, +0 for Terra.
- **Terra-written book:** no gain for Luna; +1 for Terra.
- **Own-book Terra:** 9.67× the bill for three more validation solves.

Each cell is 40 validationX problems; costs include failures and exclude book preparation.

What the Luna / Terra experiment changes

Stronger solver: yes, more coverage

- **Without ACE:** Terra 29/40; Luna 24/40.
- **Cost matters:** Terra uses a \$1.00 cap and recorded tariffs ×10.

Stronger writer: no clear unlock

- **Luna + Terra book:** 24/40, matching no book.
- **Terra + own book:** only 29 → 30/40.

Training success vs book usefulness

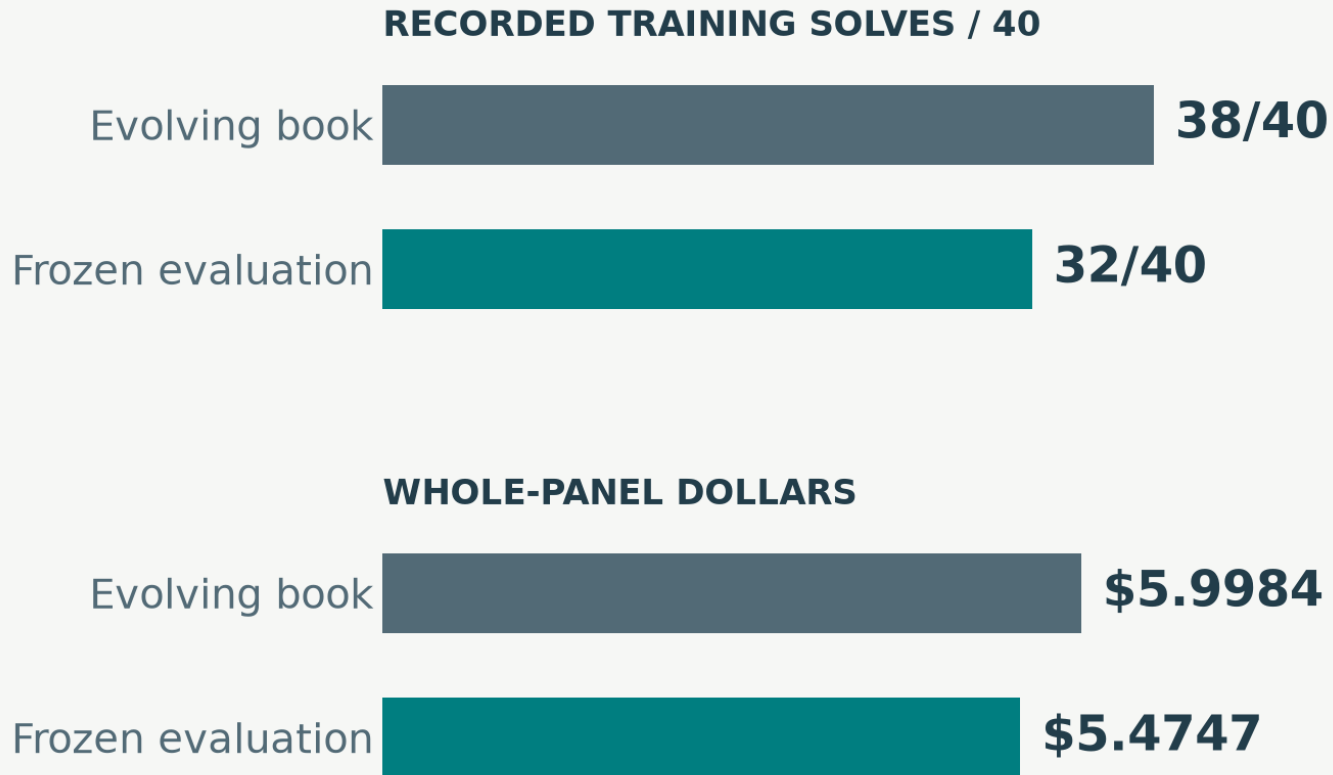
- **Generator:** Terra 38/40; Luna 33/40.
- **Transfer still needs** valid, relevant advice that the solver uses.

Next inference to test

- **Target exposed bottlenecks** in evidence or completion.
- **Larger models and more writing** do not guarantee an ACE gain.

The study does not establish that model scale unlocks ACE; an intrinsic model ceiling also remains unproven.

Yes: Terra achieved 38/40 during book creation



- **38/40 on trainX:** the book evolved between batches.
- **Generator bill:** \$5.9984 for all 40 tasks.
- **Full preparation:** \$8.0992, including writers and embeddings.
- **Frozen book:** trainX **32/40**; validationX **30/40**.

38 → 32 compares two training executions; 30/40 is a different partition, not another measured decline.

Why did 38 become 32? What we know—and do not

Known change · request admission

- **Training:** 32 requests; stop after a budget crossing.
- **Evaluation:** up to 64; reservations can deny the next call.

Known change · context

- **Training:** evolving book; older proof loop.
- **Evaluation:** frozen book; focused queries and bounded verifier feedback.

Known issue · completion is fragile

- **Unfinished progress:** time limits, parsing and final submission matter.
- **More nominal requests** do not guarantee useful admitted work.

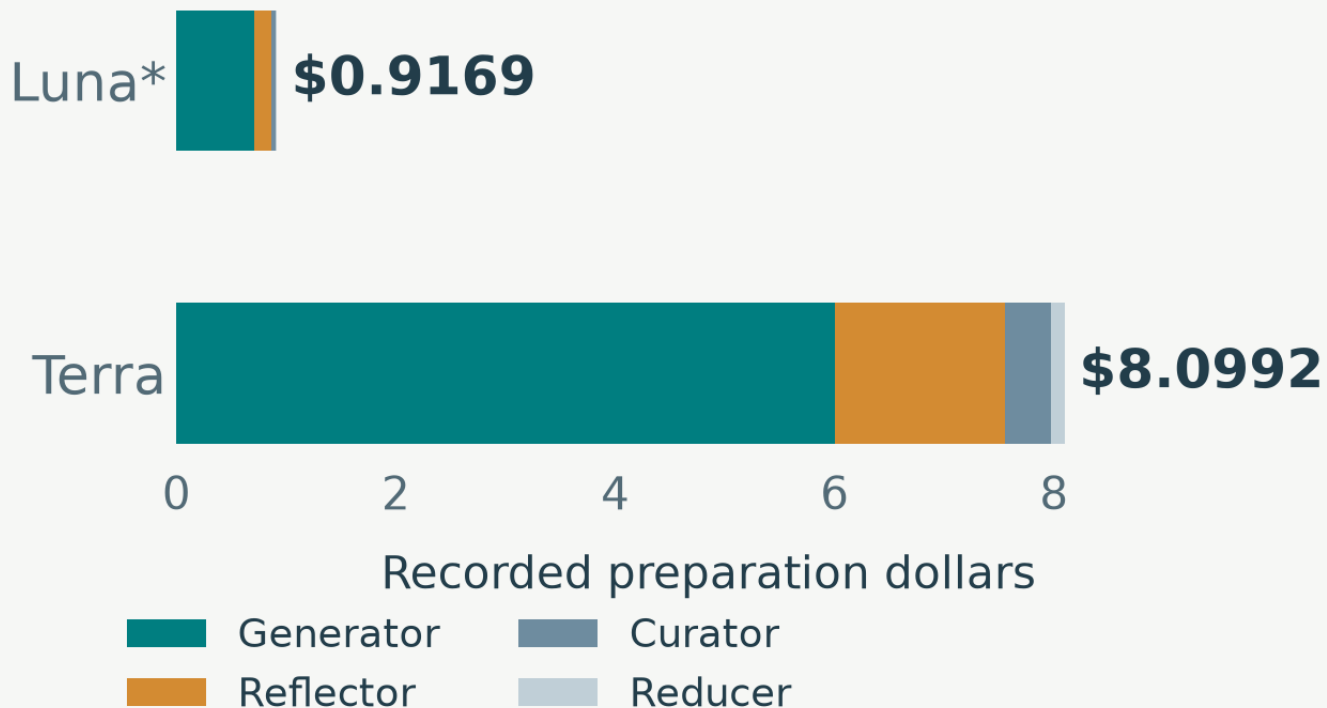
Unknown · the causal breakdown

- **Different sampled trajectories:** this is not a paired controller ablation.
- **Six net solves lost $\neq$ six proven reservation failures.**

The gap motivates diagnosis; it does not show that the finished book destroyed six proofs.

Creating a book is a real up-front expense

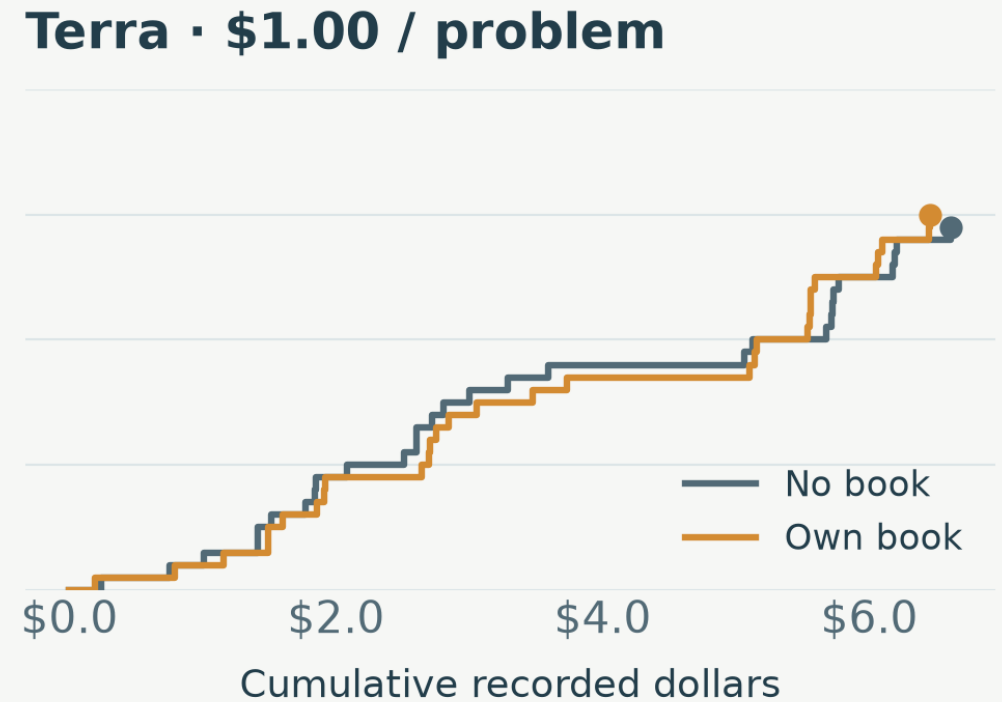
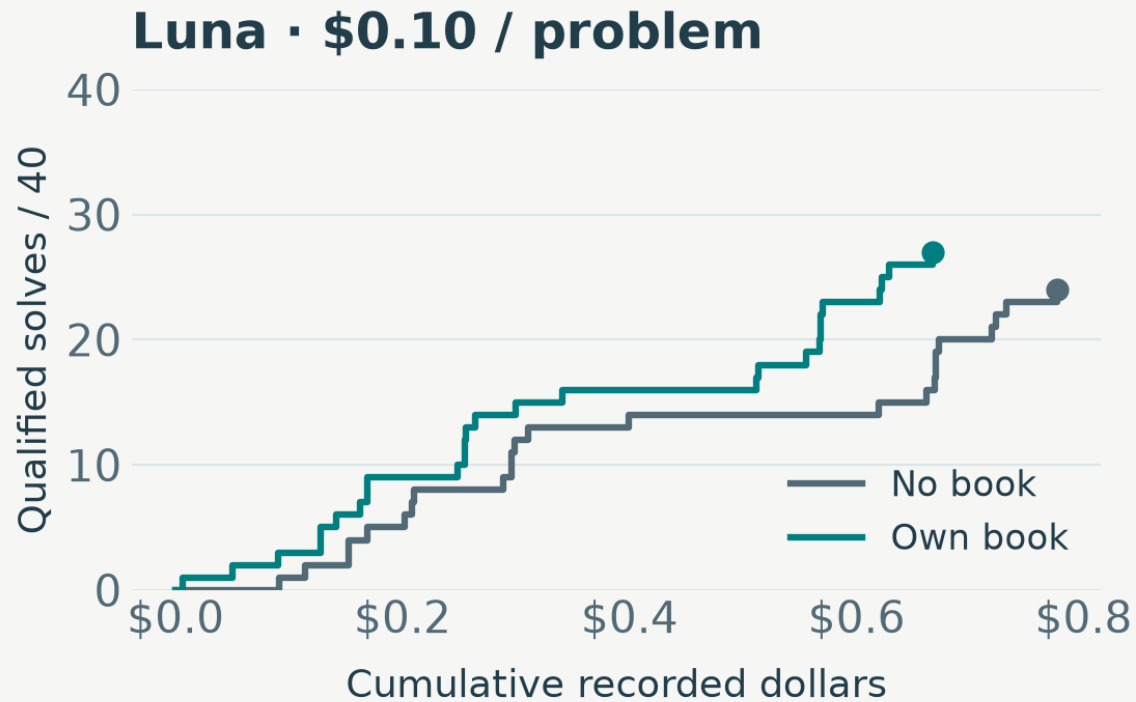
Generation is the largest component



- **Luna:** $\geq \$0.9169$; historical embeddings unknown.
- **Terra:** \$8.0992; **8.83×** Luna's known cost.
- **26 bullets each;** below the 4,000-token guard.
- **Payback requires** useful lessons to survive reduction.

Preparation and inference are separate bills; generation is the largest preparation component.

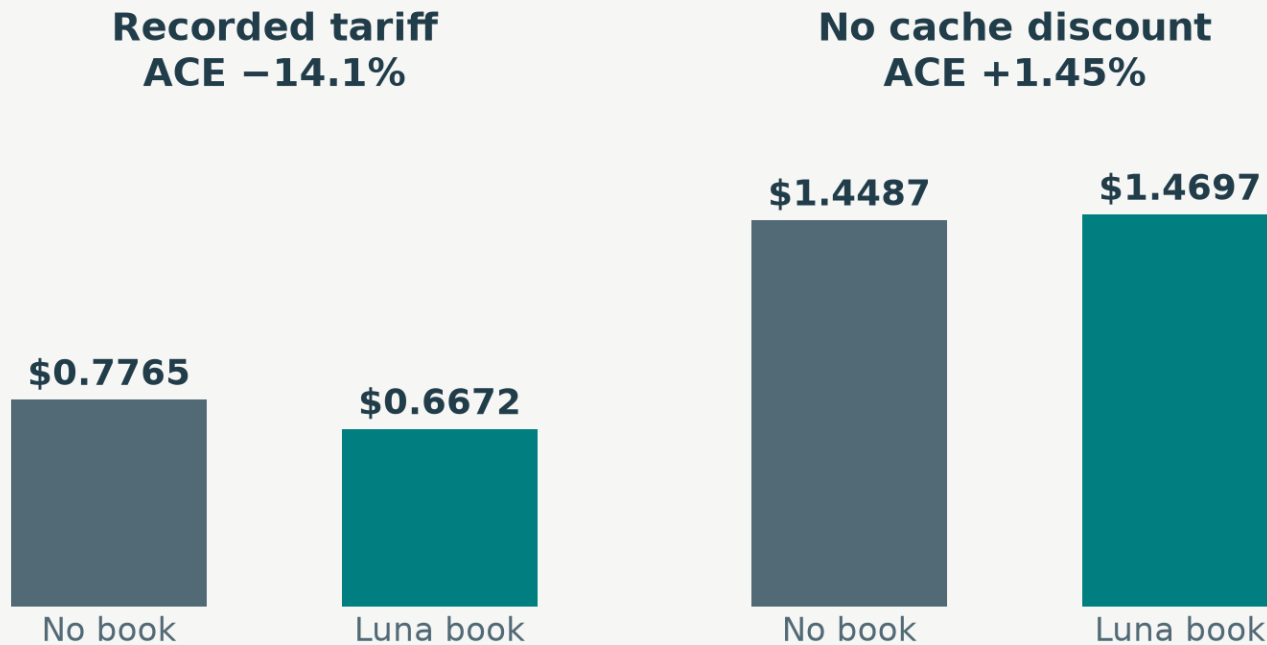
Spent versus solved: the actual recorded staircase



- **Right without up:** money spent without a solve. **Up:** a qualified proof.
- **Fixed task order; all 40 costs.** Accounting paths, not simulated smaller-cap runs.

Luna endpoints: 24 / \$0.7765 vs 27 / \$0.6672. Terra: 29 / \$6.6106 vs 30 / \$6.4513.

Cache discounts change the economic conclusion



- **Observed Luna bill:** ACE saves **14.1%**.
- **No cache discount:** the same tokens cost **1.45% more** with ACE.
- **Cached share:** 64.6% → **75.9%**; token use rises slightly.
- **Terra:** own-book uncached sensitivity is **+8.51%**.

The saving is real at the observed tariff; it cannot all be attributed to less computation.

Four failure mechanisms behind weak transfer

1 · The wrong successful ending

- **Accepted ending: nia.** Missing from the last proposed script.
- **Risk:** teach a different action from the one that closed the proof.

2 · Invalid syntax in a valid idea

- **Reducer:** valid assert becomes rejected have syntax around `Rle_0_sqr`.
- **Name lookup is insufficient:** verify the snippet in its source context.

3 · A mechanism that never arrives

- **51 parse failures; 33 duplicates.**
- **Concatenated final messages** break parsing; evaluate the fix separately.

4 · Shorter context, longer search

- **Compaction:** repeated tool calls; sometimes no proof submission.
- **Local savings** can lose useful memory and completion opportunity.

Capture → validate → retain → expose → use → finish. A failure anywhere can erase the benefit.

More allowance recovered Terra proofs, then stalled

	Original allowance	2× money	Also 2× requests / time
Luna · no ACE	4 / 8	4 / 8	4 / 8
Luna · book*	3 / 8	3 / 8	3 / 8
Terra · no ACE	4 / 8	6 / 8	6 / 8
Terra · book*	3 / 8	7 / 8	7 / 8

Caps: Luna \$0.10 → \$0.20 → \$0.20; Terra \$1 → \$2 → \$2. *Corrected own-book context.

- **Same paid prefixes:** eight selected trainX problems; no fresh restarts.
- **Extra money:** Luna unchanged; Terra gains **+2 / +4** solves.
- **Then no extra solves:** 11 residual paths face money denial; one carries an API failure.

The final plateau is not evidence of an unrestricted model-capability ceiling.

13 September: the first completion fixes did not win

Treatment	trainX	validationX	Validation bill
Historical reference	28/40	27/40	\$0.6672
A · output cap 4,096	30/40	26/40	\$1.1281
B · corrected continuation	25/40	23/40	\$0.7379

- **A exposed more search, but did not improve validation economics.**
- **Both gates failed:** no selected follow-up; neither becomes the default.
- **Cost:** \$3.6136 of a \$30 ceiling; Luna / medium / \$0.10 per problem.

A reduces the response cap from 32,768 to 4,096 tokens. Completed cycle; ongoing code work is unscored.

Are failed ideas inherited by later experiments?

Stored in Git

- **Keep results, costs, books and candidate code.**
- **A research commit does not establish a solver improvement.**

Activated by configuration

- **Opt-in:** compaction, exploration, repair and continuation.
- **The next control retains the reference;** losing flags stay off.

What really is inherited

- **The accepted controller** and its cost / coverage compromise.
- **Shared code changes** can affect several configurations.

Remaining protection to strengthen

- **One explicit reference constructor;** preserve prompt identity.
- **Separate archive, fix and promotion;** check shared behavior.

Optional treatments are isolated by switches; shared-code behavior still deserves regression checks.

The thesis claim and the advisor decisions

Claim supported by the work

- **ACE in Delphyne:** explicit artifacts and verifier feedback.
- **Useful cost / coverage control;** concrete failure diagnoses.

Claim that stays open

- **A general, reproducible ACE coverage gain.**
- **An intrinsic Luna ceiling** or a general stronger-writer advantage.

Where further work earns its cost

- **Faithful, executable evidence.**
- **Local progress must become an accepted proof** within the full budget.

Decisions for the meeting

- **Which cost / coverage trade-off serves the thesis?**
- **What evidence is still needed?** Use trainX / validationX; testX stays closed.

Evaluate improvements by completed proofs and the whole bill, with mechanism exposure recorded.

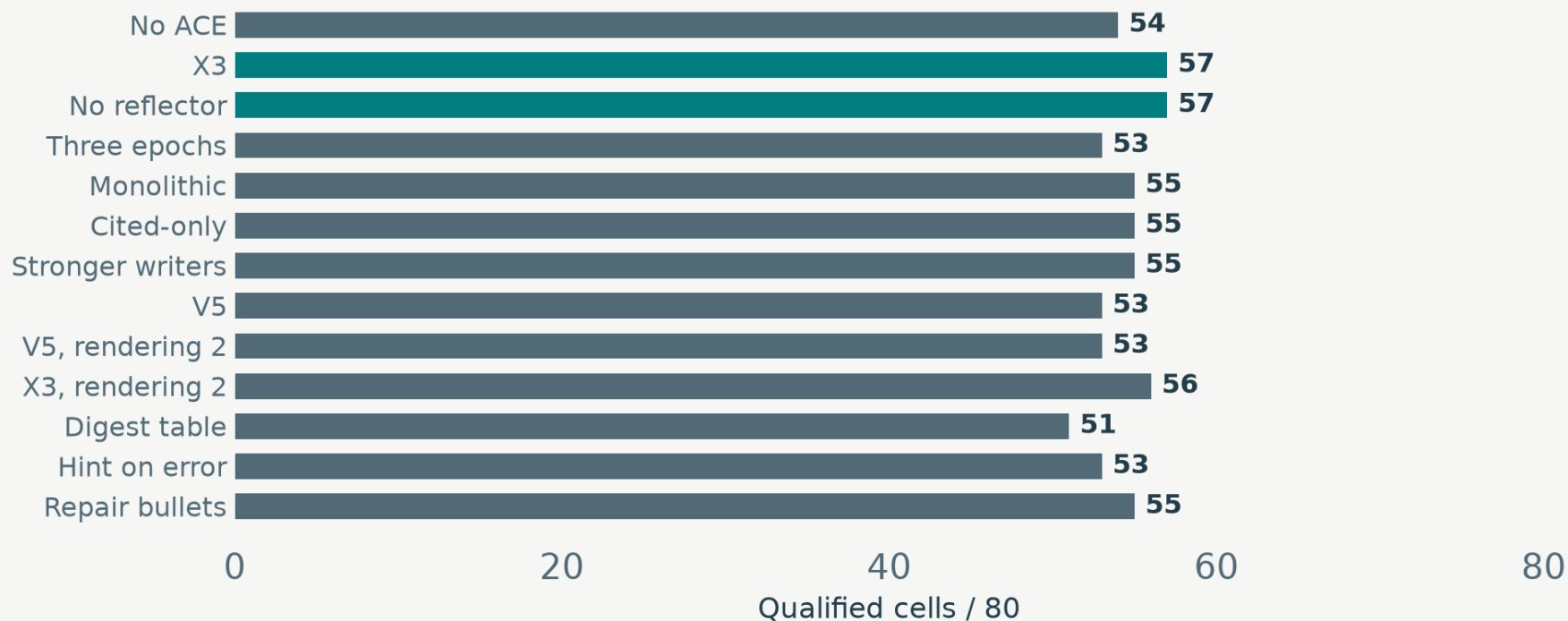
What are the exact crossover costs and caps?

Solver	Book	Solves	Panel bill	Bill / solve
Luna	None	24/40	\$0.776512	\$0.032355
Luna	Luna	27/40	\$0.667228	\$0.024712
Luna	Terra	24/40	\$0.734194	\$0.030591
Terra	None	29/40	\$6.610626	\$0.227953
Terra	Luna	29/40	\$6.852274	\$0.236285
Terra	Terra	30/40	\$6.451344	\$0.215045

- **One row = 40 validationX problems.** Recorded inference spending includes failures.
- **Preparation excluded;** historical controls reused; not 240 independent theorems.

Luna / medium / \$0.10 cap; Terra / medium / \$1.00 cap; 64 requests and 300 verifier seconds.

Which historical books improved coverage?



- **X3 and no-Reflector:** 57/80 versus no ACE **54/80**; differences remain modest.
- **Repair bullets:** 56 raw successes; **55 qualify** under the retrospective cost rule.

Two replicate identifiers on 40 validationX problems; separate books and delivery methods.

How can a run stop while money remains?

\$0.085742

Remaining balance

\$0.086368

Required reservation

DENIED

\$0.000626 over balance

- **Reserve input and maximum output charges** before dispatch.
- **Bound above balance → deny.** The hypothetical actual bill is unknown.
- **A smaller cap also constrains reasoning;** the 4,096-token trial did not win.

Unused money is not automatically spendable; looser admission must still respect financial protection.

When would preparing the book pay for itself?

Quantity	Luna	Terra
Known preparation bill	$\geq$ \$0.9169	\$8.0992
Inference saving / 40 tasks	\$0.1093	\$0.1593
Conditional break-even workload	$\approx$ 336 tasks	$\approx$ 2,034 tasks

- **Assumptions:** unchanged task mix, coverage, prices, cache reuse and usefulness.
- **Denominator:** future attempted problems; unsuccessful attempts also count.
- **Not included:** engineering / assistant costs; Luna’s missing historical embeddings.

This is arithmetic under fixed assumptions, not a prediction of external generalization.

Which evidence should an ACE lesson retain?

Proposal and execution

- **The exact submitted script** and the prefix actually executed.
- **Accepted terminal proof**, including automatic assistance such as nia.

The concrete witness

- **amc12_2000_p6**: `final_check.success = true`.
- **nia. accepted; missing from proposal**. The proof succeeded.

From source to lesson

- **Context, bindings and checked snippet** attached to the lesson.
- **Preserve valid syntax** through reflection, curation and reduction.

From lesson to benefit

- **Show delivery and use** in the later operation.
- **Then measure full proofs and total spending**; local validity is insufficient.

File to inspect: `report/ace_retrospective/data/creator_terminal_evidence.json`, entry 0.

Why is +3 solves not yet a supported gain?

What the three gains say

- **3 gains, 0 losses:** paired exact $p = 0.25$.
- **No proof of a general effect;** also no proof that ACE has zero value.

What counts as evidence

- **All expected cells;** platform failures retained in the denominator.
- **Whole-panel costs,** including failed attempts, retries and known liabilities.

What is independent

- **Cluster related theorem families and repeated seeds.**
- **validationX is reused development data;** selection is not confirmation.

What permits a practical decision

- **Support:** two-sided $p < 0.10$; 90% intervals.
- **Expansion, trade-off and promotion** are separate decisions; preserve gates.

Do not buy extra seeds merely to cross a significance threshold; testX stays closed.

Which files should I rehearse before presenting?

Question	File · relative to examples/omphalos	Search / inspect
How is ACE wired?	prove_ace.py	prove_theorem_ace; ReflectOnTrajectory
How is the book stored?	ace/ace_playbook.py; ace/ace_store.py	merge; refine; PlaybookStore.put
Where is 38/40?	experiments/campaigns/ ace_attribution_20260912/RESULTS.md	Playbook creation; Generator solves
What stopped a call?	runtime/campaign_budget.py	estimate_budget; Ledger.reserve
Terminal witness?	report/ace_retrospective/data/ creator_terminal_evidence.json	entry 0; final_check; last_tactic

Read-only rehearsal; no experiment launch, partition loader or live model call is needed.